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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Societal Project Synopsis

Name of Guide : Prof. Pramila S **Academic year- 2026-27**

Section :D **Subject Code: 1BSOP308**

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Project Title:	**"Abnormal Sensor Detection in Framework of Sewage Treatment Plants".**
Abstract	This project develops an interpretable AI-based anomaly detection system for monitoring multiple Sewage Treatment Plants (STPs) in Karnataka. It analyzes sensor data such as pH, temperature, dissolved oxygen, turbidity, and flow rate to identify abnormal readings. The system explains why a reading is considered abnormal and helps detect possible sensor faults or treatment issues. The approach is evaluated using performance metrics to improve real-time monitoring, maintenance, and STP performance**.**
Introduction	Sewage Treatment Plants (STPs) are essential for treating wastewater and reducing environmental pollution. They use different sensors to monitor important parameters such as pH, turbidity, temperature, dissolved oxygen, and flow rate.Sometimes, sensors may give abnormal or incorrect readings due to sensor faults, calibration errors, communication problems, or sudden changes in wastewater conditions. Detecting these abnormal readings manually can take time and may lead to incorrect decisions.The proposed project, "AI-Based Detection of Abnormal Sensor Readings in Karnataka Sewage Treatment Plants," uses Artificial Intelligence (AI) and Machine Learning (ML) to analyze sensor data and identify unusual patterns automatically.The system can help detect sensor problems early, improve data accuracy, reduce manual monitoring, and support efficient operation of STPs in Karnataka.

<table>
<tr><td>Existing System</td><td>
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<tr><th>Paper Details</th><th>Observations</th><th>Gap Identified</th></tr>
<tr><td>“Enhanced data-driven monitoring of wastewater treatment plants using the Kolmogorov–Smirnov test”,2024</td><td>Uses PCA–KS testing for unsupervised sensor-fault detection.</td><td>Requires further validation with real-time plant data and complex faults.</td></tr>
<tr><td>“Dual-stage soft sensor-based fault reconstruction and effluent prediction toward a sustainable wastewater treatment plant using attention fusion deep learning model”.2025</td><td>Reconstructs faulty sensor data and predicts effluent quality using deep learning.</td><td>Needs testing across different WWTPs and operating conditions.</td></tr>
<tr><td>“Enhancing fault detection in wastewater treatment plants: a multi-scale principal component analysis approach with the Kantorovich distance”.2025</td><td>Detects sensor faults under noisy conditions using an unsupervised method</td><td>Requires validation with real plant data and more fault types</td></tr>
<tr><td>“Application of AI-based techniques for anomaly management in wastewater treatment plants: A review”.2025</td><td>Reviews AI methods for anomaly detection, diagnosis, and fault management.</td><td>Challenges remain in explainability, cost, data quality, and model transferability.</td></tr>
<tr><td>“Wastewater treatment monitoring: Fault detection in sensors using transductive learning and improved reinforcement learning”.2025</td><td>Uses transductive and reinforcement learning for rare sensor-fault detection.</td><td>Class imbalance and limited fault data affect model generalisation.</td></tr>
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<tr><td>Problem Statement</td><td>Sewage Treatment Plants (STPs) use sensors to monitor parameters such as pH, temperature, turbidity, dissolved oxygen, and flow rate. However, sensors may produce abnormal or incorrect readings due to faults, environmental conditions, or maintenance issues. Manual monitoring makes it difficult to identify these problems quickly. Therefore, there is a need for an AI-based system that automatically detects abnormal sensor readings, identifies possible anomalies, and generates alerts to support timely action and improve the reliability of sewage treatment plants in Karnataka.</td></tr>
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Objectives	To develop an AI-based anomaly detection system that: 1. To Detects abnormal sensor readings. 2. To Identifies the type of abnormality. 3. To Explains why the reading is abnormal. 4. To Give a likely reason or recommendation.
Proposed System	We propose an explainable AI system for sewage treatment plants that detects abnormal sensor readings and explains their causes in simple language. The system will analyze pH, temperature, dissolved oxygen, turbidity, and flow data using threshold rules, machine-learning techniques, and cross-sensor comparisons. It will classify abnormalities such as spikes, drift, frozen readings, missing data, and process inconsistencies. For every detected anomaly, the system will provide the reason, probable cause, severity, and recommended operator action.
Methodology	1. .Data Collection: Collect sensor readings from sewage treatment plants or a suitable dataset containing parameters such as pH, turbidity, temperature, dissolved oxygen (DO), TSS, COD and flow rate. 2. Data Preprocessing: Clean the dataset by handling missing values, duplicate records and inconsistent readings. Normalize the numerical data if required. 3. Data Analysis: Analyze the sensor data to understand normal patterns, ranges and relationships between different parameters. Feature Selection Select the important sensor parameters that are useful for identifying abnormal readings. 3. AI Model Training:; Train an anomaly-detection model such as Isolation Forest, Autoencoder or One-Class SVM using the processed sensor data. 5.Anomaly Detection: Provide new sensor readings to the trained model. The model identifies whether each reading is Normal or Abnormal. 6.Model Evaluation: Evaluate the model using appropriate performance measures such as precision, recall, F1-score and accuracy when labelled data is available.

	7.Alert Generation: When an abnormal reading is detected, the system generates an alert so that the operator can inspect the sensor or treatment process. 8.Visualization: Display sensor readings, detected anomalies and alerts through a web-based dashboard.
System Requirement Specifications • **Software Requirement**	• Software Requirement: 1. Python - Main programming language 2. VS Code - Code editor for writing programs 3. Pandas & NumPy - Tools for organizing and cleaning data 4. Scikit-learn - Library for detecting abnormal patterns 5. MySQL or SQLite - Database to store sensor data 6. Flask - Tool to create a web dashboard 7. Matplotlib - Tool to draw graphs and charts 8. MQTT - Protocol for receiving sensor data 9. Grafana - Dashboard to display real-time monitoring 10. Email service - For sending alerts to operators
Applications of the Project	1. STP Monitoring: Continuously monitors sensor readings in sewage treatment plants. 2. Fault Detection: Identifies abnormal or faulty sensor readings automatically. 3. Early Warning: Alerts operators when unusual sensor behavior is detected. 4. Data Quality Improvement: Helps maintain accurate and reliable sensor data.
Conclusion	This project proposes an AI-based system for detecting abnormal sensor readings in Karnataka sewage treatment plants. The system automatically collects data from multiple sensors, identifies anomalies using machine learning, classifies the type of abnormality, and explains possible causes. It provides real-time alerts and recommended actions to plant operators through a web dashboard. This approach reduces manual monitoring, improves operational efficiency, enables timely maintenance, and ensures better water quality management. The integration of IoT sensors with AI makes the system intelligent, reliable, and scalable for implementation across multiple sewage treatment plants in Karnataka.

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SDG goals that are mapped with this project	1. SDG 6 – Clean Water and Sanitation 2. SDG 11 – Sustainable Cities and Communities 3. SDG 12 – Responsible Consumption and Production 4. SDG 13 – Climate Action 5. SDG 15 – Life on Land

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